P1: The Stationary-String Framework — A Unified FENE-P Model for Galactic, Hadronic and Laboratory Phenomenology

(A Degrees-of-Freedom Ledger, a Fragility Map, and a Forensic Audit of Its Own Errors)

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October 20, 2026

Abstract

This paper presents the FENE-P flux-string framework developed across Theorems I–X and Grand Proof I–III as a *methods* paper that lets the framework score its own claims. Its locked invariants ($\Lambda^{(0)}_{\mathrm{Meral}} = 0.07226943267767906$, $\mu_s^{\mathrm{Meral}} = 0.20440882367556615$, $\mathcal{C}_{\mathrm{univ}} = 1.6260379365521303$, $S_{\mathrm{loss}} = 19.75114251524387$, $\mathcal{I}_{\mathrm{shield}} = 2\sqrt{2}$) follow from closed forms and numerical integrals; the verification infrastructure comprises 38 automated tests and seven independent cross-check scripts (94 independent checks in total, none of which imports the solver). Our central contribution is a *degrees-of-freedom ledger* (Table 1): for each of 17 published comparisons it states the observable, the literature value, the model value, the free parameters entering, whether those parameters were fitted to that datum, and a fragility measure. The ledger shows that of the 17 comparisons only 2 are fits, 10 are transfers of the locked core, 2 are genuine predictions and 2 are structural; grouping correlated rows leaves 8 independent observable groups against 9 free degrees of freedom, i.e. an evidence ratio of 0.889. The fragility map further shows that the glueball mean is unstable: a 1% shift of $\mathcal{C}_{\mathrm{univ}}$ moves it by 27.8%, so the 1.56% agreement is *not* robust. A forensic audit record is finally given, listing eight error classes found and corrected within this work (a wrong self-adjoint extension, transposed source formulas, literature bands that turned out to be a *ghost parameter*, an H_0 dependence, a tautological lower bound, and others). The framework is therefore presented not as a proven theory but as a quantitatively auditable and self-limiting unified parametrisation.

1 Scope and the limits of the claims

This paper deliberately separates three levels of claim:

1. **Algebraic and numerical claims (exact):** the closed-form invariants, the self-adjoint extensions of the FENE-P string, the $p = 2.0000$ convergence order of the cell-centred finite-volume discretisation, and the analytic Temple–Kato enclosure. These are delivered together with the verification infrastructure.

2. **Parametrisation claims (not derived):** the transport of locks such as $\mathcal{C}_{\text{univ}}$ to galactic, Yang–Mills and laboratory observables. These involve ansatz choices, and the ledger (Table 1) states which row depends on which choice.

3. **Out of scope:** the claim that the Yang–Mills mass gap is fully solved, or that dark matter is replaced. This paper makes neither claim.

2 The locked invariants

The closed-form global-integrity invariant (a GL character integral, Euler-derived) and the dimensionless coupling, saturation ratio and shield invariant derived from it are

$$\Lambda^{(0)}_{\text{Meral}} = \frac{28}{79} \cdot \frac{1}{2}\left[\frac{163}{578} + \frac{64\sqrt{17}}{4913}\ln\left(\frac{21+5\sqrt{17}}{4}\right)\right] = 0.07226943267767906, \tag{1}$$

$$\begin{aligned} \mu_s^{\text{Meral}} &= 2\sqrt{2}\,\Lambda^{(0)}_{\text{Meral}} = 0.20440882367556615, \\ \mathcal{C}_{\text{univ}} &= \frac{F_{\text{base}}}{F_{\text{eff}}} = 1.6260379365521303, \\ S_{\text{loss}} &= \frac{e^{2\sqrt{2}}}{1-\eta_{\mathcal{T}_0}} = 19.75114251524387. \end{aligned} \tag{2}$$

$$\mathcal{I}_{\text{shield}} = \ln[S_{\text{loss}}(1-\eta_{\mathcal{T}_0})] = 2\sqrt{2}. \tag{3}$$

These numbers carry the framework's "no new calibration constant" claim: no external parameter enters the core.

3 Verification infrastructure

The solver `solver_engine.py` contains 38 automated tests (T1–T37). In addition, seven independent scripts re-derive the same physical constants without ever importing the solver: `mini_check_viii` (8/8), `mini_check_ix` (15/15), `mini_check_x` (15/15), `mini_check_gp1` (15/15), `mini_check_gp2` (17/17), `mini_check_gp3` (15/15) and the series-wide `mini_check_audit` (9/9) — 94 independent checks in total. The audit script

re-derives $\mathcal{C}_{\rm univ}$ with an ODE integration instead of the Gauss–Legendre quadrature and finds agreement to 5.6×10^{-13}; the identity $\Pi_{\rm Meral} = \mu_s^{\rm Meral}$ is confirmed to 5.6×10^{-17}.

4 The degrees-of-freedom ledger

Table 1 lists every published comparison against four questions. The *evidence class* means: FIT (a parameter was chosen to reproduce this datum), FIT-LIKE (a multiplicative factor was added after a deviation was seen), TRANSFER (the locked core is carried over from another domain), GENUINE (a discriminating prediction with no free parameter) and STRUCTURAL (a rule rather than a number).

Table 1: **Degrees-of-freedom ledger: evidence class and deviation for 17 published comparisons.** Rows are cross-checked against the JSON emitted by `ledger.py` (tolerance 2×10^{-3}).

Id	Observable	Literature	Model	Dev.	Class
T15a	a1S/h (1.03 T) [kHz]	$1.4204048 \times 10^6 \pm 5.6$	1.4204048×10^6	0.00%	Fit
T15b	a-b-g (1.03 T) [g]	0.75 ± 0.16	0.75	0.00%	Fit
T19	read-out frequency (993 mT) [kHz]	453	453	0.00%	Transfer
T22	Ps band budget ratio R	3.3387	3.3387	0.00%	Genuine
T25	loss suppression factor	31.4	31.41	0.03%	Transfer
T29	$a_{\rm Meral}$ [m/s^2]	$1.20 \times 10^{-10} \pm 2.8 \times 10^{-12}$	1.1466×10^{-10}	4.45%	Transfer
T29b	BTFR log-log slope	4	4	—	Structural
T30	5-galaxy mean deviation [%]	target 0	4.5375	4.54%	Transfer
T31	$d_{\rm ring}/\theta_g$	10 ± 0.4	10.002	0.02%	Transfer
T32	$\Delta m_{\rm gap}/\sqrt{\sigma}$	3.405 ± 0.021	3.3914	0.40%	Fit-like
T33	4-glueball mean deviation [%]	target 0	1.5563	1.56%	Fit-like
T34a	Cornell coefficient	0.295 ± 0.010	0.2985	1.18%	Transfer
T34b	$c\bar{c}$ $2S-1S$ [MeV]	589.19	596.89	1.31%	Transfer
T34c	$b\bar{b}$ $2S-1S$ [MeV]	562.96	522.25	7.23%	Transfer
T36	$\Delta\nu/\nu$ (2004 event)	2.08×10^{-6}	2.0711×10^{-6}	0.43%	Transfer
T35	resistance envelope lower edge	12	12.147	1.22%	Transfer
T37	$O_{\rm exact}$ band ratio	0.08414	0.084136	0.01%	Genuine

4.1 Three results the ledger makes explicit

(i) The evidence ratio. Grouping rows that share a knob set leaves 8 *independent* observable groups, against 7 ansatz plus 2 fit parameters, i.e. 9 free degrees of freedom. The evidence ratio is therefore $8/9 = 0.889$. The framework cannot carry a predictive burden in its present form: the agreement is a natural consequence of the model's flexibility, and the ledger says so explicitly.

(ii) The fragility map. The most sensitive row is T33: a 1% shift of the locked $\mathcal{C}_{\rm univ}$ moves the mean glueball deviation by 27.8%, because the 0^{*++} factor multiplies $\mathcal{C}_{\rm univ}$ while the gap scale does as well. The 1.56% agreement is therefore *not* robust. By contrast T34b (charmonium, 0.05%) and T36 (Vela, 0.17%) are practically stable, because $\mu_s^{\rm Meral}$ is fixed to 14 digits.

(iii) Only two genuine predictions. Of 17 comparisons, only two (T22 and T37) contain no free parameter and are genuinely discriminating; both live on the laboratory side. The remaining fifteen are fits or transfers. The same table therefore exposes the strongest and the weakest parts of the series at once.

5 Forensic audit: error classes found and fixed

The second contribution of this work is the record of the framework's own errors.

1. **Wrong self-adjoint extension (T32).** The $u = \sqrt{x}\psi$ transformation selected an incorrect extension at the critical $-1/(4x^2)$ coupling, violating the variational upper bound ($5.9158 > 5.6468$). Replacing it with a cell-centred finite-volume scheme gives $p = 1.99995$ and $\lambda_0^{\mathrm{cont}} = 5.6102039939$; an independent shooting integration gives 5.6102039940.

2. **Tautological lower bound (T32).** The Temple–Kato lower bound had been derived from the numerical matrix' own eigenvector. It was replaced by an analytic, grid-independent enclosure, $\lambda_0 \in [5.4807, 5.7674]$.

3. **Transposed source formulas (T37).** The published form of `arXiv:2605.21289` Eq. (12) and the substituted variant had been interchanged; corrected to 159.37 and 9.31 respectively.

4. **A literature band that was a ghost parameter (T35).** The bands $r_{\mathrm{ve}}/r_{\mathrm{N}} \in [1.55, 1.70]$ and $\Delta h_{\mathrm{ve}}/\Delta h_{\mathrm{N}} \in [0.30, 0.35]$ entered this work without literature provenance — git archaeology shows they arrived with the present series — and do not occur in the cited source; the claim was retracted.

5. H_0 **dependence (T29).** a_{Meral} is directly proportional to H_0; the comparison was reframed as a test of the dimensionless ratio.

6. **Post-hoc multiplicative factors (T32/T33).** The chiral dressing $(1 + \Lambda^{(0)}_{\mathrm{Meral}}/3)$ and the topological 0^{-+} term were added after deviations were observed; both are labelled as extra model interpretations.

7. **A model contradicting its own source (T35).** The single-mode laminar ansatz could not reproduce the measured flow resistance; a two-regime barrier bridge was constructed, and the fact that this construction was made *after* the measurement is stated explicitly.

8. **Hard-coded "independent" checks.** Four mini scripts copied $\mathcal{C}_{\mathrm{univ}}$ as a constant; an independent re-derivation (`mini_check_audit`) was added.

6 What would upgrade the claims

The predictive value of the framework grows only if the ansatz choices are derived. Three concrete tasks follow: (i) derive the effective potential V_{eff} from the flux-string dynamics instead of postulating it; (ii) derive the resistance/core-broadening link through the locked S_{loss}; (iii) produce at least three quantitative predictions in domains never used to fit (for example kaon spectra, or transient superfluid cores). The third is the kind of output the field rewards.

7 Reproducibility

The code lives with Theorems I–X in the `stationary-strings-v` repository (under `duran_sicimler_sim/`). Running `python solver_engine.py` executes the full test suite; `python ledger.py` regenerates the degrees-of-freedom ledger, cross-checks it against the solver output at 10^{-9} tolerance and verifies the table in this paper against the regenerated values; `mini_check_audit.py` re-derives the locked chain independently. All packages are released under CC BY 4.0.

References

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